A list of Supplementary Materials

Figure S1. Functional enrichment and PPI network analysis of DEmRNAs.

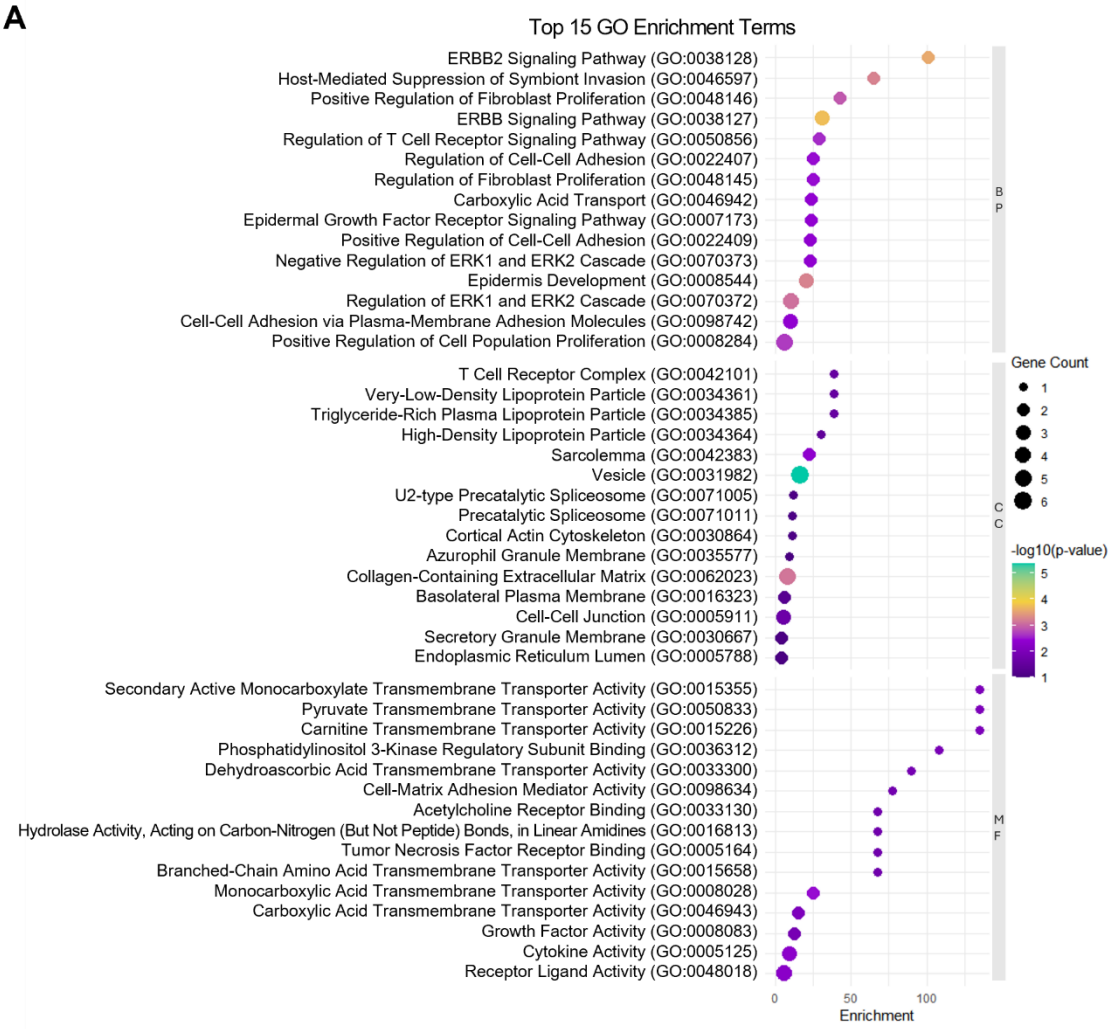
Figure S2. Kaplan−Meier overall survival analysis of selected DERNAs for all PDAC stages**.**

Figure S3. Kaplan−Meier survival analysis of the three selected mRNAs as determined using gene chip microarray data.

Figure S4. Diagnostic evaluation of FAM83A and hsa-miR-375.

Table SI. Diagnostic performance of hub ceRNA axis members based on ROC curve analysis in PDAC.

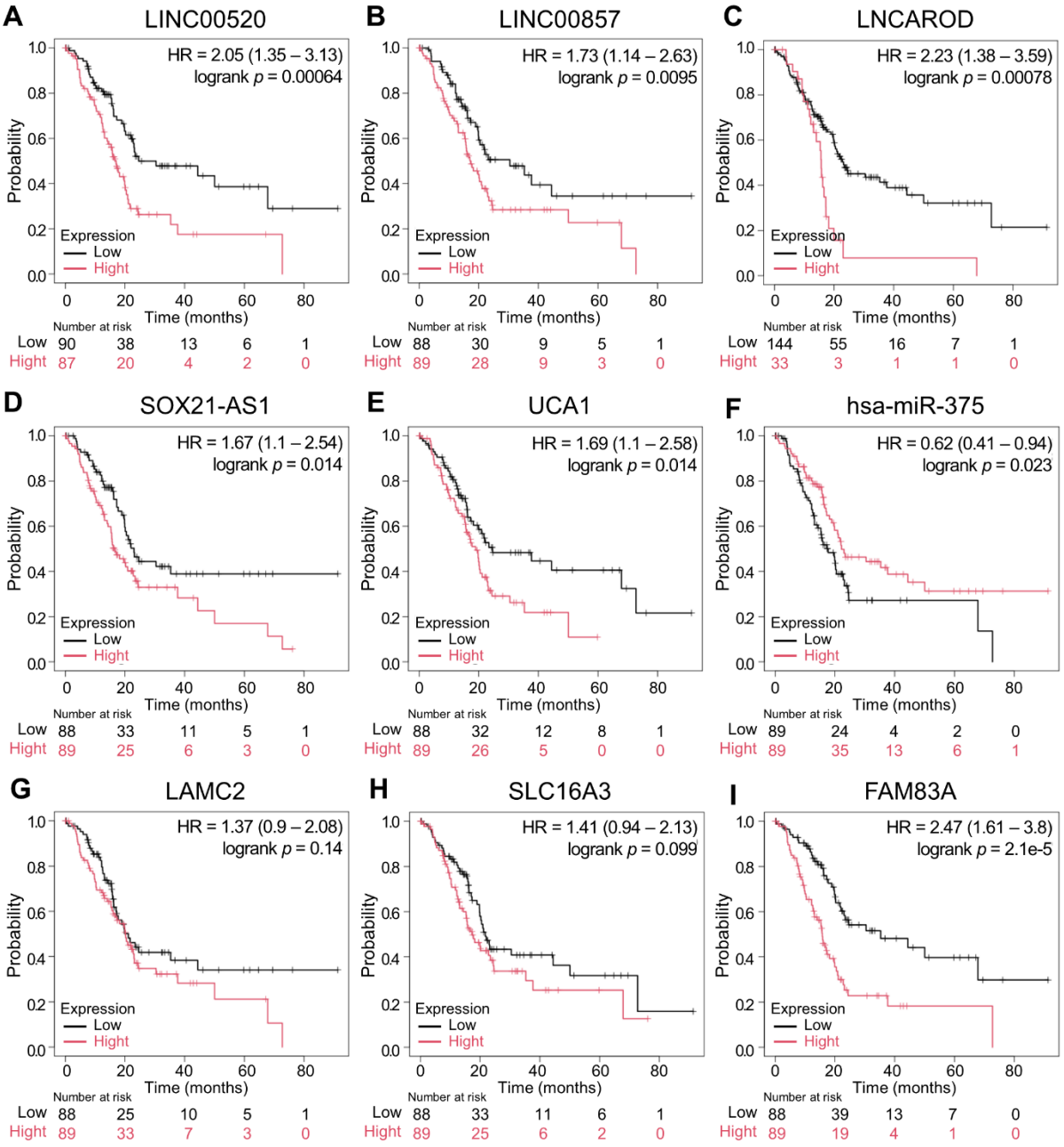
Supplementary Figure 1. Functional enrichment and PPI analysis of DEmRNAs. (A) The top 15 significantly enriched GO biological processes, cellular components, and molecular functions associated with the 38 DEmRNAs ranked by adjusted *p*-value. (B) The top 15 KEGG pathways mainly related to the 38 DEmRNAs ranked by adjusted *p*-value. (C) The PPI network constructed using the STRING database with an interaction confidence score of > 0.4, showing genes that occupy central network positions.


A
Top 15 GO Enrichment Terms
ERBB2 Signaling Pathway (GO:0038128)
Host-Mediated Suppression of Symbiont Invasion (GO:0046597)
Positive Regulation of Fibroblast Proliferation (GO:0048146)
ERBB Signaling Pathway (GO:0038127)
Regulation of T Cell Receptor Signaling Pathway (GO:0050856)
Regulation of Cell-Cell Adhesion (GO:0022407)
Regulation of Fibroblast Proliferation (GO:0048145)
Carboxylic Acid Transport (GO:0046942)
Epidermal Growth Factor Receptor Signaling Pathway (GO:0007173)
Positive Regulation of Cell-Cell Adhesion (GO:0022409)
Negative Regulation of ERK1 and ERK2 Cascade (GO:0070373)
Epidermis Development (GO:0008544)
Regulation of ERK1 and ERK2 Cascade (GO:0070372)
Cell-Cell Adhesion via Plasma-Membrane Adhesion Molecules (GO:0098742)
Positive Regulation of Cell Population Proliferation (GO:0008284)
T Cell Receptor Complex (GO:0042101)
Very-Low-Density Lipoprotein Particle (GO:0034361)
Triglyceride-Rich Plasma Lipoprotein Particle (GO:0034385)
High-Density Lipoprotein Particle (GO:0034364)
Sarcolemma (GO:0042383)
Vesicle (GO:0031982)
U2-type Precatalytic Spliceosome (GO:0071005)
Precatalytic Spliceosome (GO:0071011)
Cortical Actin Cytoskeleton (GO:0030864)
Azurophil Granule Membrane (GO:0035577)
Collagen-Containing Extracellular Matrix (GO:0062023)
Basolateral Plasma Membrane (GO:0016323)
Cell-Cell Junction (GO:0005911)
Secretory Granule Membrane (GO:0030667)
Endoplasmic Reticulum Lumen (GO:0005788)
Secondary Active Monocarboxylate Transmembrane Transporter Activity (GO:0015355)
Pyruvate Transmembrane Transporter Activity (GO:0050833)
Carnitine Transmembrane Transporter Activity (GO:0015226)
Phosphatidylinositol 3-Kinase Regulatory Subunit Binding (GO:0036312)
Dehydroascorbic Acid Transmembrane Transporter Activity (GO:0033300)
Cell-Matrix Adhesion Mediator Activity (GO:0098634)
Acetylcholine Receptor Binding (GO:0033130)
Hydrolase Activity, Acting on Carbon-Nitrogen (But Not Peptide) Bonds, in Linear Amidines (GO:0016813)
Tumor Necrosis Factor Receptor Binding (GO:0005164)
Branched-Chain Amino Acid Transmembrane Transporter Activity (GO:0015658)
Monocarboxylic Acid Transmembrane Transporter Activity (GO:0008028)
Carboxylic Acid Transmembrane Transporter Activity (GO:0046943)
Growth Factor Activity (GO:0008083)
Cytokine Activity (GO:0005125)
Receptor Ligand Activity (GO:0048018)
BP
CC
MF
Gene Count
1
2
3
4
5
6
-log10(p-value)
5
4
3
2
1
0
50
100
Enrichment
B
Top 15 KEGG Pathways
Neomycin, kanamycin and gentamicin biosynthesis
Caffeine metabolism
ECM-receptor interaction
Small cell lung cancer
Central carbon metabolism in cancer
Amoebiasis
Toxoplasmosis
Insulin secretion
Salivary secretion
Rheumatoid arthritis
Focal adhesion
Protein digestion and absorption
Human papillomavirus infection
PI3K-Akt signaling pathway
Pathways in cancer
Gene Count
1
2
3
4
5
6
P-value
0.016
0.012
0.008
0.004
50
100
Odds Ratio
C
ITGA2
B3GNT3
LAMA3
CDH3
MUC4
LAMB3
LAMC2
CEACAM6
COL17A1
CEACAM1
SLC2A1
SLC16A3
WNT2
PSCA
DKK1
CLDN18
HK2
TNFSF11
CTSE

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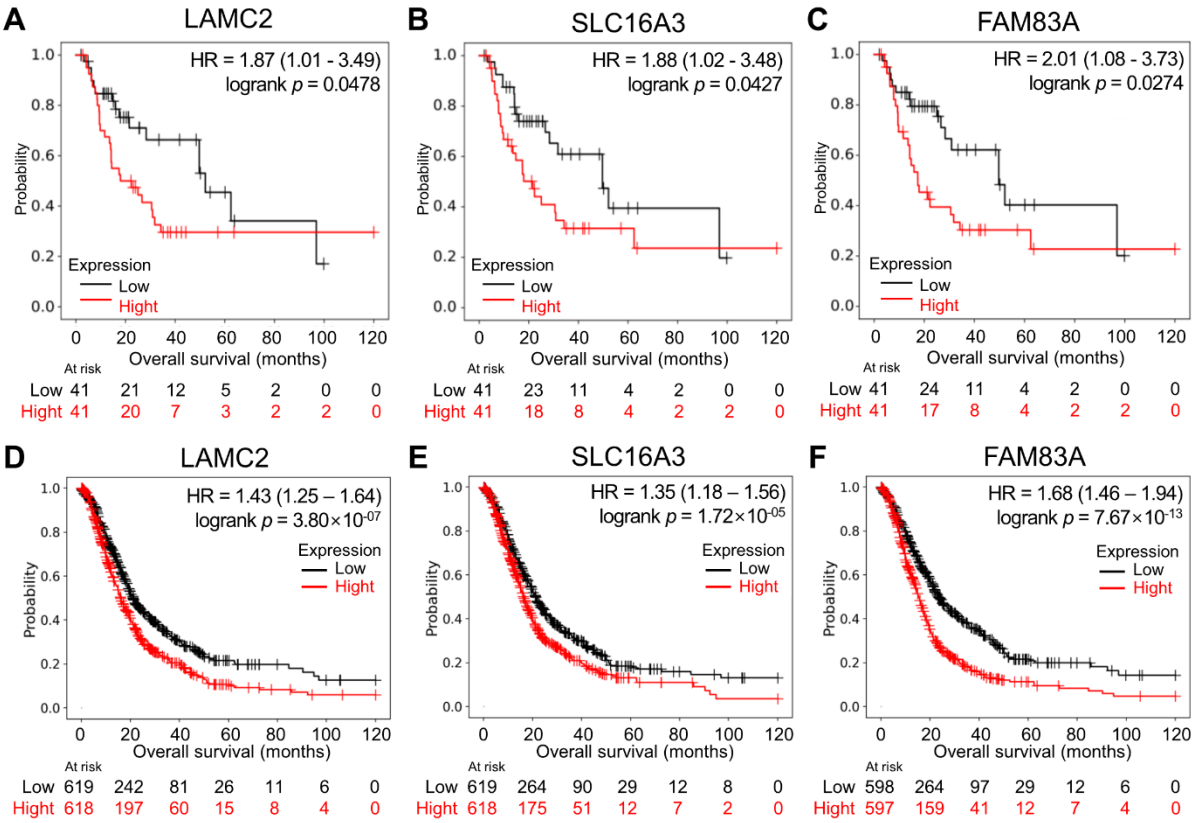
Supplementary Figure S2. Kaplan−Meier overall survival analysis of selected DERNAs across all PDAC stages. (A–E) Survival curves for the five DElncRNAs (LINC00520, LINC00857, LNCAROD, SOX21-AS1, and UCA1). (F) Survival curve for hsa-miR-375. (G–I) Survival curves for three DEmRNAs (LAMC2, SLC16A3, and FAM83A). Patients were divided into high- and low-expression groups based on median expression. Elevated DElncRNA and DEmRNA expressions were associated with shorter overall survival, whereas elevated hsa-miR-375 expression was correlated with better prognosis.


A
LINC00520
HR = 2.05 (1.35 – 3.13)
logrank p = 0.00064
B
LINC00857
HR = 1.73 (1.14 – 2.63)
logrank p = 0.0095
C
LNCAROD
HR = 2.23 (1.38 – 3.59)
logrank p = 0.00078
D
SOX21-AS1
HR = 1.67 (1.1 – 2.54)
logrank p = 0.014
E
UCA1
HR = 1.69 (1.1 – 2.58)
logrank p = 0.014
F
hsa-miR-375
HR = 0.62 (0.41 – 0.94)
logrank p = 0.023
G
LAMC2
HR = 1.37 (0.9 – 2.08)
logrank p = 0.14
H
SLC16A3
HR = 1.41 (0.94 – 2.13)
logrank p = 0.099
I
FAM83A
HR = 2.47 (1.61 – 3.8)
logrank p = 2.1e-5
Probability
Time (months)
Expression
Low
Hight
Number at risk
Low 90 38 13 6 1
Hight 87 20 4 2 0
Low 88 30 9 5 1
Hight 89 28 9 3 0
Low 144 55 16 7 1
Hight 33 3 1 1 0
Low 88 33 11 5 1
Hight 89 25 6 3 0
Low 88 32 12 8 1
Hight 89 26 5 0 0
Low 89 24 4 2 0
Hight 89 35 13 6 1
Low 88 25 10 5 1
Hight 89 33 7 3 0
Low 88 33 11 6 1
Hight 89 25 6 2 0
Low 88 39 13 7 0
Hight 89 19 4 1 0

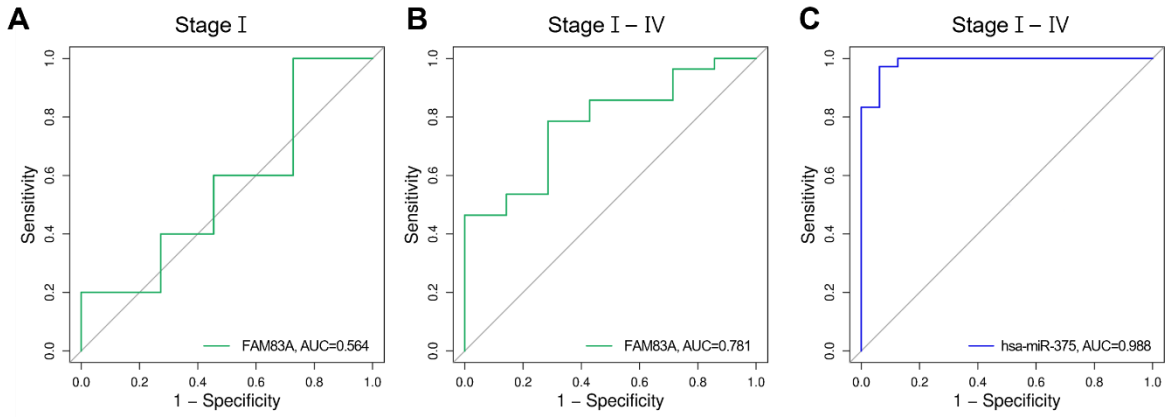
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Supplementary Figure S3. Kaplan−Meier survival analysis of the three selected mRNAs as determined using gene chip microarray data. (A–C) Overall survival curves for LAMC2, SLC16A3, and FAM83A in stage I patient tissues. (D–F) Overall survival curves for LAMC2, SLC16A3, and FAM83A in all-stage patient tissues. Patients were stratified into high- and low-expression groups based on median expression, and for each gene, higher expression was associated with poorer overall survival.



Supplementary Figure S4. Diagnostic evaluation of FAM83A and hsa-miR-375. (A) ROC curve for FAM83A in stage I patients, showing limited discriminative ability due to many missing values in the proteomic dataset. (B) ROC curve for FAM83A across all stages, demonstrating modest diagnostic performance. (C) ROC curve for hsa-miR-375 across all stages, showing potent discriminative ability.



Supplementary Table SI. Diagnostic performance of key ceRNA axis members based on ROC curve analysis in PDAC.

Dataset / Gene	AUC	95% CI	Sensitivity (%)	Specificity (%)	*p*-value
Transcriptome (Stage I)					
LINC00857	0.969	0.92-1.00	87	100	2.17e-72
UCA1	0.859	0.74-0.98	83	90	3.42e-09
LAMC2	0.959	0.90-1.00	96	95	3.26e-49
SLC16A3	0.928	0.85-1.00	78	100	5.02e-30
FAM83A	0.944	0.87-1.00	96	90	6.72e-34
Transcriptome (Stage I-IV)					
LINC00857	0.971	0.95-1.00	91	95	6.37e-321
UCA1	0.824	0.74-0.91	76	90	6.48e-13
LAMC2	0.968	0.94-1.00	92	95	5.63e-247
SLC16A3	0.967	0.94-0.99	86	100	8.75e-287
FAM83A	0.943	0.89-1.00	94	90	6.92e-52
has-miR-375	0.988	0.97-1.00	97	94	1.23e-391
Proteome (Stage I)					
LAMC2	0.967	0.92-1.00	94	91	4.01e-102
SLC16A3	0.922	0.86-0.99	94	80	2.93e-36
FAM83A	0.564	0.46-0.88	55	83	1.06e-01
Proteome (Stage I-IV)					
LAMC2	0.982	0.96-1.00	98	91	6.9e-678
SLC16A3	0.925	0.88-0.97	91	95	7.84e-90
FAM83A	0.781	0.56-0.82	65	70	3.50e-03